

A Battery-Powered Wild Animal Tracking Device Using a PTZ Camera and Deep Learning

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Abstract—In this paper, we propose a battery-powered wild animal tracking device using a Pan-Tilt-Zoom (PTZ) camera and deep learning. The proposed tracking device detects wild animals using YOLOv5 and tracks the detected wild animals using DeepSort. In addition, the proposed tracking device realizes tracking for a wide range by controlling a PTZ camera according to the movement direction of the detected wild animals. In the experiment, we developed a prototype for the proposed tracking device and conducted the field test of the developed prototype. We confirmed and discussed cases where the developed prototype could and could not track wild animals. The energy consumption of the developed prototype during tracking was 526.03J at the daytime and 730.99J at the nighttime.

I. INTRODUCTION

Recently, accidents and crop damage caused by wild animals such as bears are significant problems in Japan. According to a report by the Ministry of the Environment [1], the number of accidents caused by wild bears was 220 in 2023. That is about three times the number of accidents in 2022. To reduce accidents, early detection and tracking of wild animals are important issues.

To automatically detect wild animals, wild animal detection devices were developed in [2]–[4]. These devices take pictures using a fixed camera and detect wild animals using deep learning. If wild animals are detected, these devices generate an alarm and notify people of the detection information using email and web applications. However, these devices do not track the detected wild animals. If we can know the movement direction of the detected wild animals, it may reduce accidents caused by wild animals by notifying people of the movement information.

As traditional wild animal tracking devices, tracking devices attached to wild animals were developed in [5]–[7]. [5] tracks wild animals by sending the position information of wild animals obtained from Global Positioning System (GPS) through Wireless Sensor Networks. Similarly, devices based on Radio Frequency IDentification (RFID) [6] and Low Power Wide Area (LPWA) [7] were developed. However, attaching such devices to wild animals is hard and dangerous because we must capture wild animals once.

On the other hand, tracking devices using a fixed camera and deep learning were developed in [8]–[10]. [8] and [9] track animals by detecting animal positions using deep learning and comparing the detected positions between frames. [10] tracks animals by predicting animal poses using deep learning and comparing the predicted poses between frames. However, the tracking is limited within the angle of view

of the camera because the camera direction is fixed. If the camera direction can be controlled, we can track animals for a wide range.

In this paper, we propose a battery-powered wild animal tracking device using a Pan-Tilt-Zoom (PTZ) camera and deep learning. The proposed tracking device detects wild animals using YOLOv5 [11], which is one of the object detection models in deep learning. In addition, the proposed tracking device assigns identifiers to the detected wild animals using DeepSort [12], which is one of the multiple object tracking algorithms. Moreover, the proposed tracking device tracks the detected wild animals by controlling the PTZ camera according to the movement direction of the detected wild animals. We implement all functions such as object detection, object tracking, PTZ camera control, etc., on a single microcomputer board at the edge side.

The contributions of this paper are as follows:

- Realization of automatic tracking of wild animals for a wide range by using a PTZ camera and deep learning
- Realization of a battery-powered wild animal tracking device with a single microcomputer board

To track wild animals in real-time with low power consumption using a single microcomputer board, we restrict the processing of YOLOv5 and DeepSort. The proposed tracking device performs YOLOv5 only when motions are detected to reduce unnecessary power consumption. The proposed tracking device does not perform DeepSort when only one animal is detected to reduce unnecessary processing time and power consumption.

The rest of this paper is organized as follows. Section II describes related work. Section III describes YOLOv5 and DeepSort. Section IV describes the proposed wild animal tracking device. Section V describes the experimental results. Finally, section VI describes the conclusion and future work.

II. RELATED WORK

Object tracking devices using a PTZ camera and motion detections were developed in [13]–[15]. These devices detect objects using background subtraction methods. The device in [13] compares features such as positions and colors of the detected objects between frames and controls the PTZ camera direction to move the detected object to the center of the image. The devices in [14] and [15] control the PTZ camera direction when the center coordinate of the detected object changes and when the detected object moves to a specified area, respectively.

Compared with [13]–[15], where all moving objects are detected by motion detection methods, we use deep learning for object detection in this paper. Object detection models based on deep learning can infer the types and positions of objects. Therefore, the proposed tracking device tracks

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only when target wild animals are detected by deep learning. As a result, we expect to reduce the power of the proposed tracking device by avoiding unnecessary tracking.

Object tracking devices using a PTZ camera and deep learning were developed in [16] and [17]. These devices detect objects for all frames taken by the PTZ camera in which the initial direction is fixed. In [16], the device extracts object boundaries using background subtraction methods and recognizes objects using ResNet [18]. Then, the device tracks the objects using a tracking algorithm [19] based on a spatial regularization correlation filter. In [17], the device detects objects using YOLOv5 and tracks the objects using DeepSort. These devices control the PTZ camera direction to move the object to the center of the image.

Compared with [16] and [17], we propose a battery-powered wild animal tracking device in this paper. The proposed tracking device performs motion detection using motion sensors before detecting objects using deep learning. As a result, we expect to reduce the power of the proposed tracking device because the device performs deep learning only if motions are detected. In addition, the proposed tracking device realizes object detection for multiple directions by controlling the PTZ camera direction using multiple motion sensors. Moreover, the proposed tracking device decides on one tracking target for tracking when multiple animals are detected to simplify a PTZ control, which is not described in [16] and [17]. The proposed tracking device does not perform DeepSort when only one animal is detected to reduce processing time and power consumption. Therefore, the proposed tracking device achieves real-time tracking with low power compared with [16] and [17].

III. PRELIMINARIES

A. YOLOv5

To detect wild animals, the proposed tracking device uses YOLOv5 [11]. YOLOv5 is one of the object detection models with high processing speed. YOLOv5 can infer the types and positions of objects.

Figure 1 shows the architecture of YOLOv5. YOLOv5 consists of Convolution-BatchNorm-SiLu (CBS), Upsample, Concat, C3, Spatial Pyramid Pooling Fusion (SPPF), and Conv. CBS is used to generate local feature maps. Upsample is used to enlarge feature maps. Concat is used to concat feature maps. C3 is used to perform the highly accurate inference while reducing the amount of calculation. SPPF is used to provide robustness for different image sizes. Conv is used to infer a bounding box for each pixel of feature images.

B. DeepSort

DeepSort [12] is one of the multiple object tracking algorithms. DeepSort is based on Simple Online and Realtime Tracking (SORT) [20]. In addition, DeepSort uses appearance features extracted from CNNs to achieve multiple object tracking.

Figure 2 shows the overview of DeepSort. DeepSort consists of object detection, position prediction, feature extraction, and matching cascade. In the object detection, objects are detected by CNNs. In the position prediction, the positions of the detected objects are predicted by a Kalman

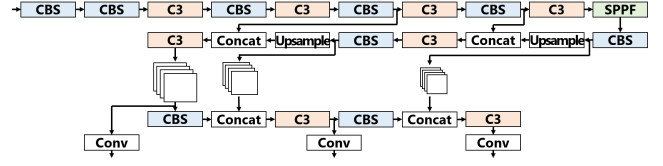


Fig. 1. Architecture of YOLOv5 [11].

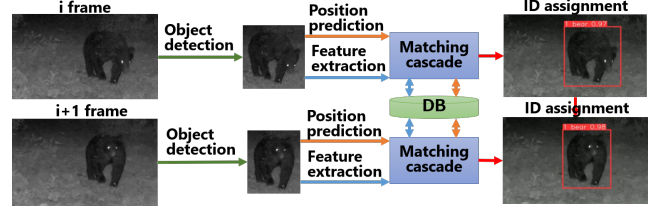


Fig. 2. Overview of DeepSort.

filter. In the feature extraction, the features of the detected objects are extracted by CNNs. In the matching cascade, identifiers are assigned to the detected objects by checking the similarities of the positions and features of the objects between frames. In this paper, the proposed tracking device uses DeepSort to deal with parents and children of wild animals and groups of wild animals.

IV. PROPOSED WILD ANIMAL TRACKING DEVICE

The proposed tracking device tracks wild animals by controlling a PTZ camera. The proposed tracking device is powered by a battery and a solar panel to be used outdoors where a power supply is not available. The proposed tracking device is based on the wild animal detection device in [2]. In this section, we describe the functional requirements, structure, and operation flow of the proposed tracking device.

A. Functional Requirements

To detect and track wild animals, the proposed tracking device has the following functions.

- PTZ camera rotation using multiple motion sensors
- Wild animal detection using deep learning for frames
- Decision of one tracking target when multiple wild animals are detected
- Control of the PTZ camera direction (zoom ratio) according to the center coordinate (area) of the tracking target
- Notification to people using sound and light and transmission of the detection information to a server

We assume that the proposed tracking device is placed near school roads, tourist roads, vegetable fields, and mountains. Figure 3 shows the overview of the proposed tracking device.

The proposed tracking device rotates the PTZ camera using multiple motion sensors to realize object detection for a wide range. We may realize object detection for a wide range by rotating the PTZ camera without using motion sensors. However, it consumes more power because the PTZ camera rotates regardless of the appearance of wild animals. To reduce the power consumption of the proposed tracking

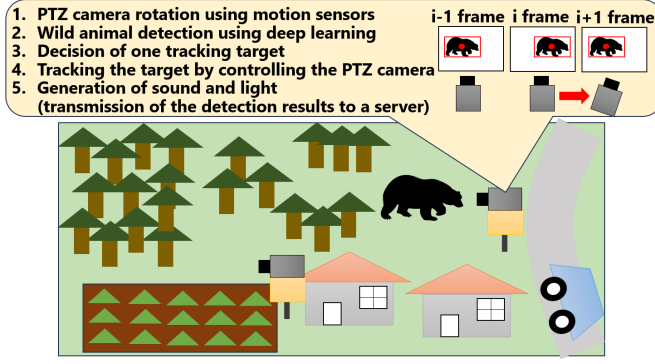


Fig. 3. Overview of the proposed tracking device.

device, we use multiple motion sensors to trigger tracking wild animals.

After rotating the PTZ camera, the proposed tracking device automatically detects target wild animals using YOLOv5. We can easily detect target wild animals because the types and positions of objects can be inferred by deep learning. In addition, the detection of various wild animals can be easily realized by changing the trained deep learning model.

When multiple wild animals are detected, the proposed tracking device decides on one tracking target. By deciding on one tracking target from the detected wild animals, we can simplify the control of the PTZ camera.

After deciding on one tracking target, the proposed tracking device tracks the target by controlling the direction and zoom ratio of the PTZ camera. For example, as shown in Fig. 3, the proposed tracking device controls the PTZ camera direction according to the movement direction of the tracking target. As a result, we can realize tracking for a wide range with a single PTZ camera.

To notify people of wild animal detections, the proposed tracking device generates sound and light. In addition, the proposed tracking device transmits the detection information to a server. Note that the generations of sound and light and the transmission of the detection information to a server are the same as [2].

B. Structure

Figure 4 shows the structure of the proposed tracking device. The proposed tracking device performs object detection, object tracking, PTZ camera control, etc., using a single microcomputer board. Three motion sensors are used to detect motions for the front, right, and left sides. The PTZ camera is used to take images and track wild animals and is connected to the microcomputer board using Wi-Fi. The speaker and light-emitting device are used to generate sound and light to warn people near wild animals. The wireless module is used to send the detection results to a server. The USB memory is used to store the detection results.

The proposed tracking device is powered by a battery and a solar panel. The battery life is important to be used outdoors where a power supply is not available. In the proposed tracking device, the infrared projector, speaker, and light-emitting device do not need to be in constant operation.

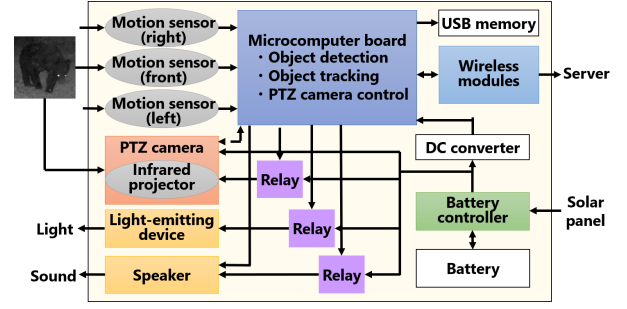


Fig. 4. Structure of the proposed tracking device.

By turning on the infrared projector for only taking images at the nighttime, we reduce unnecessary power consumption. To operate such components when the operations are needed, the relay modules are used to manage the power supplies. Note that the PTZ camera is always powered because the PTZ camera takes a long time to initialize and connect to the microcomputer board.

C. Operation Flow

Figure 5 shows the operation flow of the proposed tracking device. The dotted arrows represent the operation flow during tracking.

The proposed tracking device starts its operation when one of the motion sensors is sensed. First, the proposed tracking device checks the shooting mode of the PTZ camera. If the shooting mode is a night mode, the proposed tracking device turns on the infrared projector. Then, the proposed tracking device rotates the PTZ camera and reads multiple frames. After reading multiple frames, the proposed tracking device checks some movements between frames using a frame difference method. If some movements are checked, the proposed tracking device performs object detection using YOLOv5.

When target wild animals are not detected, the proposed tracking device controls the PTZ camera to zoom in if the PTZ camera can zoom in or to zoom out if the PTZ camera cannot zoom in. It is difficult to detect small objects with few features by object detection models such as YOLOv5. By zooming in to enlarge objects, the detection accuracy can be improved. Note that the proposed tracking device repeats the above processes until the default tracking time has elapsed.

When target wild animals are detected, the proposed tracking device tracks the detected wild animals. The proposed tracking device generates sound and light only once. While generating sound and light, the proposed tracking device decides on one tracking target and controls the PTZ camera to track the target.

Decision of one tracking target. To control the PTZ camera, the proposed tracking device decides on one tracking target when multiple wild animals are detected. The proposed tracking device assumes that the tracking target is the parent or head animal because there is a high possibility that wild animals will be moving together with the parent and head.

When multiple wild animals are detected, the proposed tracking device decides on one tracking target by referring to the identifier, height, and movement direction of the detected

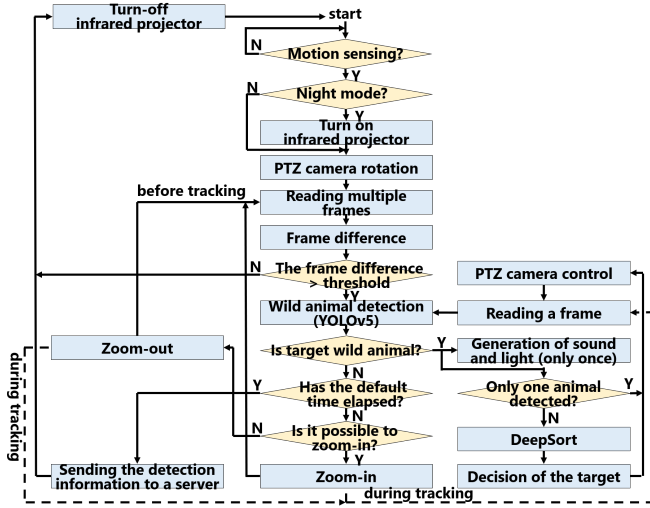


Fig. 5. Operation flow of the proposed tracking device.

wild animals. First, the proposed tracking device assigns identifiers to the detected wild animals using DeepSort. By DeepSort, the same identifier can be continued from the next frame onwards. Then, the proposed tracking device estimates the height of the detected wild animals with the same identifiers using bounding boxes and a pinhole camera model. If the difference in the height between the detected wild animals is larger than a threshold, the proposed tracking device considers the highest wild animal as the parent. If the parent cannot be tracked, the proposed tracking device tracks the head animal by estimating the movement direction of the detected wild animals using their center coordinates.

Control of the PTZ Camera. The proposed tracking device tracks the target by controlling the PTZ camera. Figure 6 shows examples of the PTZ camera controls. To simplify the PTZ camera control, the change ratio for the direction and zoom are multiple fixed values.

The proposed tracking device controls the PTZ camera in the following cases.

- Control of PTZ camera direction (zoom ratio) when the center coordinates of the target change by more than threshold values
- Control of the PTZ camera direction when the target moves to a specified area

Figure 6(a) shows an example of the PTZ camera control when the center coordinates of the target change by more than threshold values. In this example, the threshold for the Pan control is 100. The center coordinates of the target are (400,150) at the i th frame. The center coordinates of the target are (280,150) at the $i+j$ th frame. In this case, the proposed tracking device controls the PTZ camera direction to the left because the x coordinate changes by 120 (> 100).

Figure 6(b) shows an example of the PTZ camera control when the target moves to a specified area. This approach is used to avoid losing the target at the edge of the image. In this example, the green area is the specified area. The center coordinates of the target are (200,150) at the i th frame. The center coordinates of the target are (150,150) at the $i+j$ th

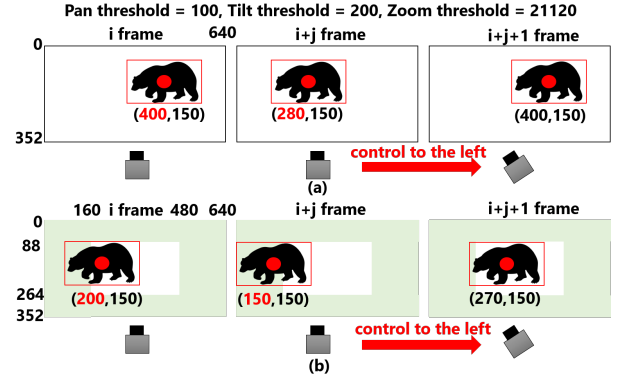


Fig. 6. Examples of the PTZ camera controls:(a) when the center coordinates of the target change and (b) when the target moves to the specified area.

frame. In this case, the proposed tracking device controls the PTZ camera direction to the left because the target moves to the specified area.

V. EXPERIMENTS

In the experiment, we develop a prototype for the proposed tracking device to demonstrate whether the proposed tracking device can detect and track wild animals. In addition, we evaluate the operation time, power consumption, and energy consumption of the developed prototype.

Figure 7 shows the developed prototype. Table I represents the specification of the developed prototype. In the field test, we used Raspberry Pi 4 (Raspberry Pi OS Bullseye 64bit) as a microcomputer board and Ctronics CTIPC-285C-Z4 (takes a 640x352 image) as a PTZ camera. We implemented a tracking program on Raspberry Pi 4 by referring to [21] and using Python 3.9. In addition, we implemented the PTZ control and zoom-in (out) using multi-threading. We set the thresholds and specified area for the PTZ control to the same values in Fig. 6. We also set the default tracking time to 60s. In the field test, sound and light were not generated and detection results were not transmitted to a server to prevent a case in which wild animals run away. We developed two prototypes and placed the developed prototypes in Aizu-Wakamatsu City.

The developed prototypes used YOLOv5n to detect wild animals and Omni-Scale Network (OSNet) [22], [23] to extract features of the detected wild animals. YOLOv5n was trained using 52,475 images for the training, 1,328 images for the validation, and 1,311 images for the test. As the training environment, we used GeForce RTX 3080 Ti, CUDA 12.0, PyTorch 2.0.0, and TensorFlow 2.10.0. We set the image size to 320x320 to reduce the inference time of YOLOv5n. The dataset included Japanese bears, wild boars, raccoon dogs, palm civets, dogs, cats, deer, antelopes, foxes, rabbits, crows, persons, and cars. The precision and recall of YOLOv5n were 0.909 and 0.789 for test images. OSNet was trained using 13,875 images for the training, 2,520 images for the validation, and 2,665 images for the test. As the training environment, we used GeForce RTX 3070, CUDA 11.7, PyTorch 1.13.1, and TensorFlow 2.9.1.

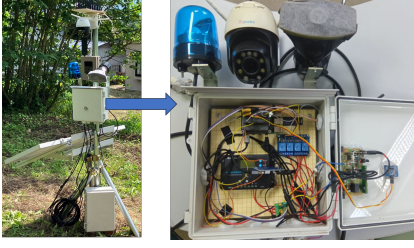


Fig. 7. Developed prototype.

TABLE I
SPECIFICATION OF THE DEVELOPED PROTOTYPE.

Name	Component
Microcomputer board	Raspberry Pi 4 Model B 4GB RAM
GPIO expansion board	GeekPi Easy Multiplexing Board
Wireless module	Candy Pi Lite+ D
Motion sensor	Seeed Studio PIR Motion Sensor
PTZ camera	Ctronics CTIPC-285C-Z4
Light-emitting device	Patlight SL08-M1JN
Speaker	DC 12V 60W 110db speaker
Relay module	SODIAL 4ch 5V relay module
Battery controller	Renogy 10A PWM Charge Controller
Battery	Long DC12V 36Ah
Solar panel	Renogy DC 12V 100W
USB memory	Elecom MF-PKU3032GBK/E USB Memory, 32 GB
DC converter	YMax Backup Converter Board BE0917

We set the image size to 64x64 to reduce the inference time of OSNet. The dataset included Japanese bears, wild boars, raccoon dogs, palm civets, cats, deer, antelopes, foxes, and rabbits. The mAP and rank-1 of OSNet were 0.771 and 0.994 for test images. In the field test, we set the tracking targets to wild animals except persons and cars. In addition, we implemented the trained models on Raspberry Pi 4 to run YOLOv5n and OsNet with tflite-runtime 2.11.0.

A. Wild Animal Tracking Results

We show three tracking results in Fig. 8. Figure 8(a) shows frames from the tracking video of wild boars in Aizu-Wakamatsu City at 1:18 a.m. on October 24, 2023. The developed prototype was tracking a wild boar, but was unable to detect the boar at the 72nd frame. The developed prototype tracked again by re-detecting the wild boar at the 102nd frame by zooming in.

Figure 8(b) shows frames from the tracking video of a raccoon dog in Aizu-Wakamatsu City at 10:14 p.m. on November 5, 2023. The center coordinates of the raccoon dog were (525,216) at the 0th frame. The center coordinates of the raccoon dog were (424,263) at the 19th frame. Since the x coordinate changed by more than the threshold (100), the developed prototype tracked the raccoon dog by controlling the PTZ camera direction to the left. On the other hand, the center coordinates of the raccoon dog were (383,313) at the 39th frame. Since the y coordinate (313) was larger than the specified y coordinate (264), the developed prototype tracked the raccoon dog by controlling the PTZ camera direction to the down.

Figure 8(c) shows frames from the tracking video of wild boars in Aizu-Wakamatsu City at 6:16 p.m. on October 18, 2023. The developed prototype controlled the PTZ camera direction to the left and down at the 44th frame to track a wild boar. However, the developed prototype lost the wild

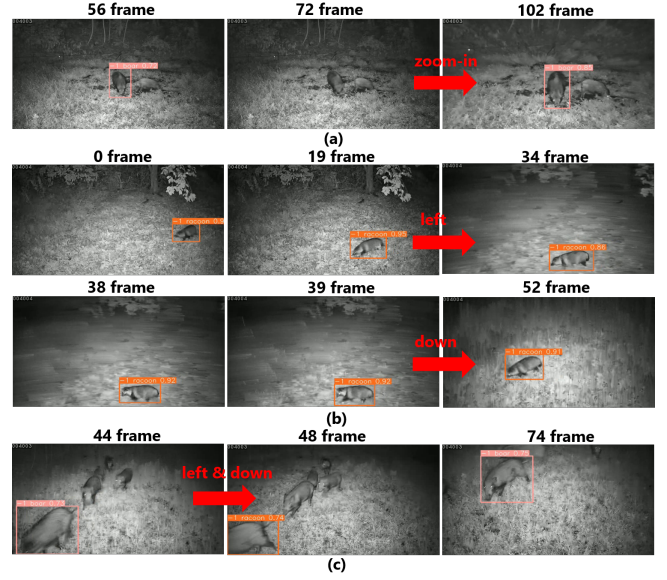


Fig. 8. Wild animal tracking examples by the developed prototype:(a) Zoom, (b) Pan-Tilt, and (C) multiple animals.

boar at the 74th frame after controlling the PTZ camera. This is because the control time of the PTZ camera does not correspond to the movement speed of the wild boar. In addition, the developed prototype did not detect all wild boars because the training data set included few images taken by the PTZ camera.

In the field test, we confirmed cases where the developed prototype could and could not track wild animals. In our future work, we are going to change the control time of the PTZ camera according to the movement speed of wild animals to improve the tracking accuracy. Moreover, we are going to predict the movement direction of wild animals to deal with the case where the animal is running or staying at the edge of the image. In addition, we are going to improve the detection accuracy by training with many images taken by the PTZ camera.

B. Operation Time, Power, and Energy

We estimated the operation time, power consumption, and energy consumption of the proposed tracking device. The operation time was obtained using perf_counter functions in the Python code to control the proposed tracking device. The power consumption was obtained by inserting INA260 [24] to the output of the battery controller. We used power values with an error within 10% of the product of current values and voltage values. The energy consumption was obtained by multiplying the operation time and power consumption. Note that we estimated the operation time, power consumption, and energy consumption when the developed prototype tracked a bear image indoor because no Japanese black bear was detected after INA260 was introduced.

Table II represents the operation time (T), power consumption (P), and energy consumption (E) of the developed prototype. *Tracking* and *Idle* represent the operation during tracking and idle, respectively. *Constant monitoring*

TABLE II
OPERATION TIME, POWER, AND ENERGY OF THE DEVELOPED
PROTOTYPE.

Process	Day			Night		
	<i>T</i> [s]	<i>P</i> [W]	<i>E</i> [J]	<i>T</i> [s]	<i>P</i> [W]	<i>E</i> [J]
PTZ camera rotation	2.60	9.69	25.19	2.59	11.75	30.43
Reading multiple frames	1.07	8.20	8.77	0.96	11.24	10.79
Frame difference	0.05	8.26	0.41	0.03	11.77	0.35
Detection before tracking	0.27	8.89	2.40	0.31	11.91	3.69
Reading a frame	0.01	9.18	0.09	0.01	12.55	0.12
Detection during tracking	0.15	9.65	1.44	0.14	13.08	1.83
PTZ camera control	0.45	9.82	4.41	0.82	13.14	10.77
Zoom-in (out)	6.62	10.22	67.65	7.11	13.26	94.27
Tracking	54.91	9.58	526.03	56.23	13.00	730.99
Idle	-	7.59	-	-	7.51	-
Constant monitoring	-	8.58	-	-	11.71	-

represents constant monitoring without using motion sensors. In *Constant monitoring*, the PTZ camera rotation, reading multiple frames, and frame difference method were performed constantly.

The operation time of the developed prototype depended on the default tracking time. Among the operation times, the zoom control took the most time. This is because the zoom control time included a zoom time and camera focusing time. In addition, the FPS of the developed prototype was 5.12 at the daytime and 5.46 at the nighttime. The FPS depended on the inference time of YOLOv5n. To improve the FPS, we are going to reduce the inference time of the YOLOv5n by parallel execution of YOLOv5n for multiple frames.

The power consumption of the developed prototype of *Tracking* was 9.58W at the daytime and 13.00W at the nighttime. The power consumption at the nighttime was higher than that at the daytime because the developed prototype turned on the infrared projector. Compared with *Idle*, the power consumption of *Tracking* increased significantly because of the high power consumption of the PTZ camera control and the inference by YOLOv5n. To extend battery life, we are going to use a low-power PTZ camera and small object detection models.

Compared with *Constant monitoring*, the developed prototype could reduce the power consumption of *Idle* by 11.53% at the daytime and 35.86% at the nighttime. This is because the developed prototype did not perform the PTZ camera rotation, reading multiple frames, and frame difference method during *Idle*. In addition, the developed prototype turned on the infrared projector only when motions were sensed at the nighttime.

The energy consumption of the developed prototype during tracking was 526.03J at the daytime and 730.99J at the nighttime. The energy consumption at the nighttime was higher than that at the daytime because of the high power consumption at the nighttime.

VI. CONCLUSION

In this paper, we proposed a battery-powered wild animal tracking device using a PTZ camera and deep learning. The proposed tracking device tracks wild animals by detecting wild animals using YOLOv5, assigning identifiers to the detected wild animals using DeepSort, and controlling the PTZ camera. In the field test, we confirmed cases where the developed prototype could and could not track wild animals. The energy consumption of the developed prototype during

tracking was 526.03J at the daytime and 730.99J at the nighttime.

In our future work, we are going to dynamically change the control time of the PTZ camera according to the movement speed of wild animals to improve the tracking accuracy. In addition, we are going to optimize the energy consumption of the proposed tracking device through the use of a low-power PTZ camera and the miniaturization and parallelization of object detection models. Moreover, we are going to improve the detection accuracy by training with many images taken by the PTZ camera.

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